

Statistical Inference and Modeling

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Semester II, 2025

- Lecture 1: Introduction
- Lecture 2: Model selection
- Lecture 3: Resampling
- Lecture 4: Linear regression
- Lecture 5: Robust regression
- Lecture 6: Classification
- Lecture 7: Logistic regression
- Lecture 8: Neural networks
- Lecture 9: K-nearest neighbor
- Lecture 10: Bayesian decision, LDA, QDA, and naive Bayes
- Lecture 11: Support vector machine
- Lecture 12: Tree-based methods
- Lecture 13: Bagging, random forest, boosting
- Lecture 14: Principal component analysis
- Lecture 15: Clustering algorithms
- Lecture 16: Gaussian mixture model clustering
- Lecture 17: DBSCAN clustering

17. Gaussian Mixture Model Clustering

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- 1 Table of contents
- 2 Supervised vs unsupervised learning
- 3 K-mean clustering algorithms (recap)
 - K-mean clustering algorithms (recap)
 - Problems with K-mean (recap)
 - GMM— solution to a K-mean's problem
- 4 Gaussian Mixture Model (GMM)
 - Settings of a Gaussian Mixture Model (GMM)
 - Deriving a probabilistic model
 - Deriving a probabilistic model for GMM
- 5 Estimating the model parameters in GMM
 - GMM parameter estimation — just the idea
 - GMM vs. K-mean
 - Warnings
 - Settings of the parameter estimation in GMMs
 - Estimating parameters using stationary conditions
 - Expectation-Maximization (EM)
 - Applying EM to GMM parameter estimation
 - GMM parameter estimation using EM — the algorithm
 - Training and inference phase of GMM clustering
 - Coding for EM in GMM clustering

Supervised vs unsupervised learning

Supervised learning

$$\{X_j, y_j\}_{j=1}^m$$

Using $X_j \rightarrow \hat{y}_j$

Unsupervised learning

$$\{X_j\}_{j=1}^m$$

Using $X_j \rightarrow$ the data behavior

Examples of unsupervised learning

- ① data clustering
- ② dimensionality reduction:
 - data visualization
 - data compression
- ③ data association (rule-based machine learning)

Characteristics of unsupervised learning

- ① No ground truth $\rightarrow$ hard to assess the result from the model.
- ② More subjective $\rightarrow$ no simple goal for the analysis $\rightarrow$ usually based on rules.

Unsupervised learning algorithms

- ① k-mean clustering
- ② hierarchical clustering
- ③ $\Rightarrow$ Gaussian Mixture Model (GMM)
- ④ $\Rightarrow$ Density-based spatial clustering of applications with noise (DBSCAN)

- 1 Table of contents
- 2 Supervised vs unsupervised learning
- 3 **K-mean clustering algorithms (recap)**
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 - Problems with K-mean (recap)
 - GMM— solution to a K-mean's problem**
- 4 Gaussian Mixture Model (GMM)
 - Settings of a Gaussian Mixture Model (GMM)
 - Deriving a probabilistic model
 - Deriving a probabilistic model for GMM
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 - Applying EM to GMM parameter estimation
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GMM parameter estimation — just the idea
GMM vs. K-mean
Warnings

K-mean clustering algorithms (recap)

Target:

$$\underset{C_1, C_2, \dots, C_K}{\operatorname{argmin}} \sum_{k=1}^K \frac{1}{|C_k|} \sum_{j \in C_k} \|\mathbf{x}_j - \bar{\mathbf{x}}_k\|_2^2$$

Algorithm

- **Initialization:**

- Assume the number of clusters (K).
- randomly assign a number of 1 to K to each training sample.

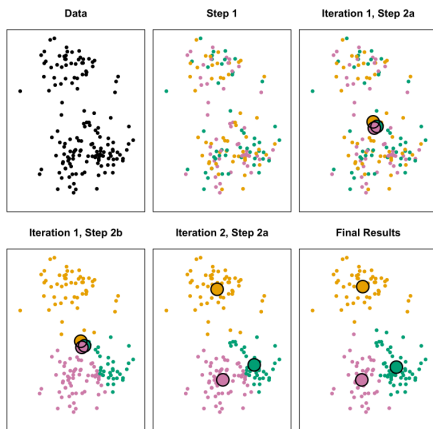
- **Start:** iterate the following steps until the cluster assignments stop changing

- 1 For each cluster, compute the cluster centroid $\bar{\mathbf{x}}_k$.

$$\bar{\mathbf{x}}_k = \frac{1}{|C_k|} \sum_{j' \in C_k} \mathbf{x}_{j'}$$

- 2 Assign each training sample to the cluster whose centroid is the closest to the training sample.

Visualization



Problems with K-mean

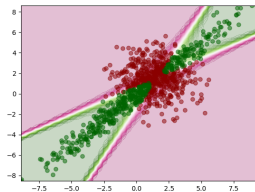
The main problem of K-mean:

- They assume that the data points belonging to one cluster will not overlap with the data points of other clusters...
⇒ making **a hard decision** to which cluster data points belong to ...
- The Euclidean distance of the data datapoints is solely responsible for the clustering...
⇒ K-mean often provides the data clusters in circular shape, thus, indicating the lack of the judgement for **data density**.
- As a result, they are likely to fail, if the data points of different densities overlap.
- Cannot handle noise in training samples. GMM cannot solve this problem as well**.

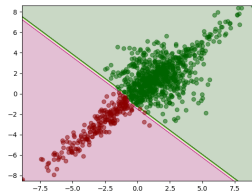
Solution:

- Enabling **soft-decision** by measuring the probability that the data belong to each cluster.
- Enabling **density estimation** by using **a probabilistic model** that can capture the data density of each cluster.

สังเกต เรากำหนด $k=2$ ขั้นตอน: Estimate $\hat{\mu}, \hat{\Sigma}$ ให้เราหาค่า Analytic



GMM

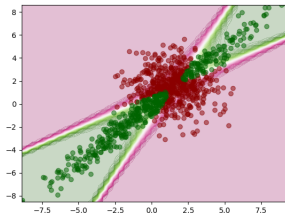


K-Mean

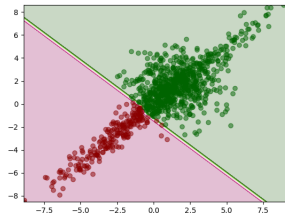
GMM— solution to a K-mean's problem

We can achieve the two goals using a probabilistic model, called **Gaussian mixture model (GMM)**:

- ⇒ GMM can be used to capture the probability of the data points belonging to each cluster.
- ⇒ The probability and the model parameters can be jointly estimated using an algorithm called **expectation maximization (EM)**. ★★★ → มีบางค่าเหมือน MLE, MAP
- ⇒ GMM clustering that will cluster the data based on **the probability density**, instead of the Euclidean distance.
- ⇒ The cluster of the data samples can be identified by **the probability where the data samples belong to a cluster**.



GMM



K-Mean

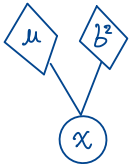
- ① Table of contents
- ② Supervised vs unsupervised learning
- ③ K-mean clustering algorithms (recap)
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 - Problems with K-mean (recap)
 - GMM— solution to a K-mean's problem
- ④ **Gaussian Mixture Model (GMM)**
 - Settings of a Gaussian Mixture Model (GMM)
 - Deriving a probabilistic model
 - Deriving a probabilistic model for GMM
- ⑤ **GMM parameter estimation — just the idea**
 - GMM vs. K-mean
 - Warnings
- ⑤ Estimating the model parameters in GMM
 - Settings of the parameter estimation in GMMs
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Graphical representation

x (circle) represent
← a random variable

μ (diamond) ← a parameter, e.g. b^2 (diamond) : a variance

Example: x is a Gaussian RV with μ, b^2



$$x \sim \mathcal{N}(\mu, b^2)$$



$$x \sim \text{Bernoulli}(p)$$

Joint probabilities

1. If x_1 and x_2 depends on each other

$(x_1) - (x_2) \quad P(x_1, x_2) = P(x_1, x_2)$

2. If x_1 and x_2 are independent of each...

$(x_1) \quad (x_2) \quad P(x_1, x_2) = P(x_1) P(x_2)$

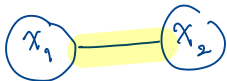
3. If x_1, x_2, x_3 are independent of each other

$(x_1) \quad (x_2) \quad (x_3) \quad P(x_1, x_2, x_3)$
 $= \underline{P(x_1) P(x_2) P(x_3)}$

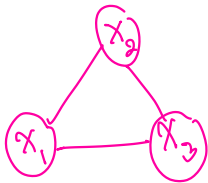
4. If x_3 is independent of x_1, x_2



$$p(x_1, x_2, x_3) = p(x_1, x_2) p(x_3)$$



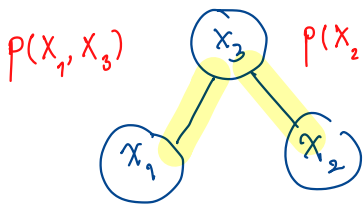
5. If x_1, x_2 , and x_3 are all related
(all depend on each other)



$$p(x_1, x_2, x_3) \\ = p(x_1, x_2, x_3)$$

What about

$$p(x|y) = \frac{p(x,y)}{p(y)}$$



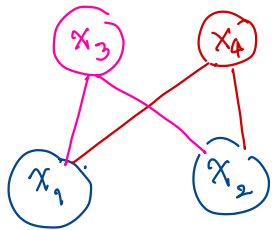
$$p(x_1, x_3)$$

$$p(x_2, x_3)$$

$$p(x_1, x_2, x_3) = ?$$

$$p(x_1 | x_3) p(x_2 | x_3) p(x_3)$$

What about



$$P(x_1, x_2) = \int_{x_3} p(x_1, x_2, x_3) dx_3$$

$$\begin{aligned} & \uparrow \\ & P(x_1, x_2, x_3) \\ &= p(x_1 | x_3) p(x_2 | x_3) p(x_3) \end{aligned}$$

$$\begin{aligned} & P(x_1, x_2, x_4) \\ &= p(x_1 | x_4) p(x_2 | x_4) p(x_4) \end{aligned}$$

$$p(x_3, x_4) = \iint_{x_1, x_2} p(x_1, x_2, x_3, x_4) dx_2 dx_1$$

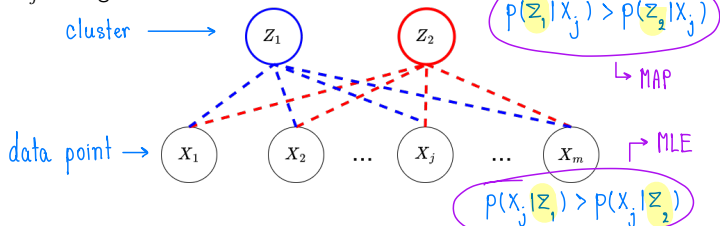
$$\begin{aligned} & \uparrow \\ & P(x_1, x_2, \underline{x_3}, \underline{x_4}) \rightarrow p(x_1 | x_3, x_4) p(x_2 | x_3, x_4) \end{aligned}$$

$$= p(x_1, x_2 | x_3, x_4) p(x_3) p(x_4)$$

Settings of a Gaussian Mixture Model (GMM)

Settings:

- Let $k = 1, \dots, K$ denotes the index of the cluster.
- Let the input features $\mathbf{x}_j \in \mathcal{R}^n$ in the training set $\{\mathbf{x}_j\}_{j=1}^m = \{\mathbf{x}_1, \dots, \mathbf{x}_m\}$ which can be partitioned into K clusters, $\{\mathbf{x}_j\}_{j=1}^m = \bigcup_{k=1}^K \{\mathbf{x}_j\}_{j \in k}$.
- Let Z be an integer random variable, i.e., $Z \in \{1, 2, \dots, K\}$, denoting the cluster that an input feature $\mathbf{x}_j$ belongs to...



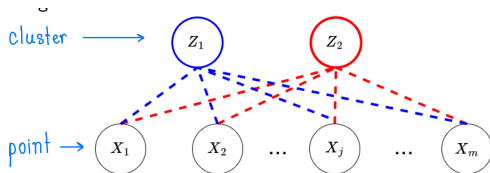
- $\Rightarrow$ The probability that the k th cluster exists is controlled by $p_Z(z = k; \pi_k)$.
- $\Rightarrow$ Also, $\sum_{k=1}^K p_Z(z = k; \pi_k) = 1$
- $\Rightarrow$ If the input features $\mathbf{x}_j$ belonging to the k th cluster, $p_{X|Z}(\mathbf{x}_j | z_k)$ should be high.

Settings of a Gaussian Mixture Model (GMM)

Derive the relationship
between RVs form $\rightarrow$

$$x_1 \perp x_2 \perp x_3 \dots \perp x_m \mid z_1, z_2$$

data point $\rightarrow$



$$p(x_1, x_2, \dots, x_m, z_1, z_2) = \prod_{i=1}^m p(x_i \mid z_1, z_2) \cdot p(z_1, z_2)$$

$$p(x_1, x_2, \dots, x_m, z_1) = \prod_{i=1}^m p(x_i \mid z_1) \cdot p(z_1) \rightarrow p(z_1) = \int p(x_1, x_2, \dots, x_m, z_1) dx$$

$\forall x = \{x_1, x_2, \dots, x_m\}$

$$p(x_1, x_2, \dots, x_m, z_2) = \prod_{i=1}^m p(x_i \mid z_2) \cdot p(z_2) \rightarrow p(z_2) = \int p(x_1, x_2, \dots, x_m, z_2) dx$$

$\forall x = \{x_1, x_2, \dots, x_m\}$

Deriving a probabilistic model

We can formulate the relationship between $z = k$ and $\mathbf{x}_j$ if $j \in k$ as follows:

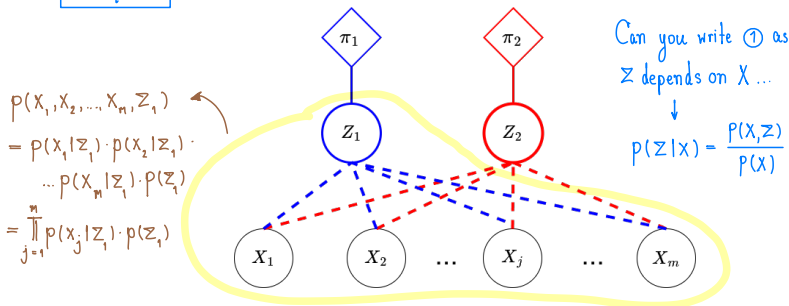
$$z \sim p_Z(z = k; \pi_k)$$

$$\mathbf{x}_j \stackrel{iid}{\sim} p_{\mathbf{X}}(\mathbf{x}_j | z = k) \text{ for } j \in k\text{th cluster}$$

With the above relationship, we can draw a graphical relationship between $\mathbf{x}_j$ and $z = k$ as the picture below, which is equivalent to the following joint probability:

$$\sum_i \pi_i = 1$$

$$p_{\mathbf{X}, Z}(\mathbf{x}_j, z = k) = p_{\mathbf{X}|Z}(\mathbf{x}_j | z = k) p_Z(z = k; \pi_k) \text{ — ①}$$



Deriving a probabilistic model

The following joint probability

$$\mathbf{x}_j = [\mathbf{x}_{ij}]_{i=1}^n$$

$$p_{\mathbf{X}, Z}(\mathbf{x}_j, z = k) = p_{\mathbf{X}|Z}(\mathbf{x}_j | z = k) p_Z(z = k; \pi_k)$$

can be marginalized to provide the probability of each training sample $\mathbf{x}_j$, which is

$$p_{\mathbf{X}}(\mathbf{x}_j) = \sum_{k=1}^K p_{\mathbf{X}, Z}(\mathbf{x}_j, z = k). \quad \text{Note } z \text{ is discrete.}$$

Then, the probability of the k th cluster given the input features can be computed from

$$p_{Z|\mathbf{X}}(z = k | \mathbf{x}_j) = \frac{p_Z(z = k; \pi_k) p_{\mathbf{X}}(\mathbf{x}_j | z = k)}{p_{\mathbf{X}}(\mathbf{x}_j)}.$$

We will apply this same concept to the Gaussian Mixture Model.

Assume $\mathbf{Z}$, $\mathbf{X}$ are independent.

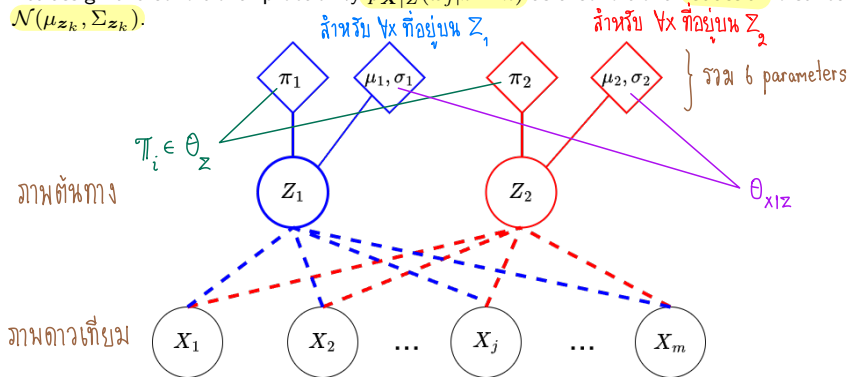
$$p_{\mathbf{Z}|\mathbf{X}}(\mathbf{z}|\mathbf{x}) = \frac{p_{\mathbf{Z}, \mathbf{X}}(\mathbf{x}, \mathbf{z})}{p_{\mathbf{X}}(\mathbf{x})}$$

Gaussian Mixture Model (GMM)

→ Statement เหมือน LDA.

GMM:

- Let $\pi_k := p_Z(z = k; \pi_k)$ be a discrete probability which is the **weight** for the conditional probability $p_{\mathbf{X}|Z}(\mathbf{x}_j|z = k)$.
- Let assign the conditional probability $p_{\mathbf{X}|Z}(\mathbf{x}_j|z = k)$ as a conditional **Gaussian** distribution $\mathcal{N}(\mu_{z_k}, \Sigma_{z_k})$.



Gaussian Mixture Model (GMM)

$$p(x) = \sum_k \overset{\text{weight}}{p(z_k)} \cdot \overset{\text{discrete}}{p(x|z_k)}$$

ดูอย่างนี้สิ

Thus, the distribution $p_{\mathbf{X}}(\mathbf{x}_j)$ can be rewritten as:

$$p_{\mathbf{X}}(\mathbf{x}_j) = \sum_{k=1}^K \pi_k p_{\mathbf{X}|Z}(\mathbf{x}_j|z = k).$$

$$\begin{aligned}\pi_k &= p(z = k) \\ \tau_{jk} &= p(z = k | \mathbf{x}_j)\end{aligned}$$

The probability of the k th cluster given the input features will be denoted as τ_{jk} , which is

$$\tau_{jk} := p_{Z|\mathbf{X}}(z = k | \mathbf{x}_j) = \frac{\pi_k p_{\mathbf{X}|Z}(\mathbf{x}_j | z = k)}{p_{\mathbf{X}}(\mathbf{x}_j)}$$

Note that a GMM involves with the following parameters: $\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K$. These parameters are necessarily for computing $p_{\mathbf{X}}(\mathbf{x}_j)$, and then, τ_{jk} .

ถ้า $\tau_{jk=2} > \tau_{jk=3}$
 $\therefore$ Sample $\mathbf{x}_j$ ต้องอยู่ Cluster 2 ไม่ใช่ 3

$$p_{\mathbf{X}|Z}(\mathbf{x} | z = k; \mu_k, \Sigma_k)$$

GMM parameter estimation — just the idea

Initialize. Randomize the model parameters of a probabilistic model.

Iterate the following two steps until convergence.

- 1 Assign the **soft-cluster memberships**, which is the probability density of data sample belonging to each cluster:

$$\tau_{jk} \leftarrow \underbrace{\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}}_{\text{weight}} \underbrace{\}_{k=1}^K$$

Initialize Phase.

where $\tau_{jk} = p_Z(z = k | \mathbf{x}_j)$.

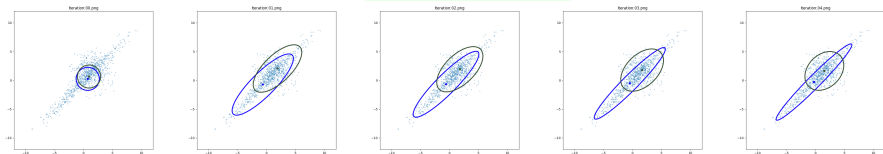
- 2 Update the model parameters of the probability density: $\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K$. These parameters can be updated using τ_{jk}

$$\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K \leftarrow \tau_{jk}$$

Loop

- τ_{jk} weakly imposes the cluster on the data samples. The update of model parameters will progress toward maximizing τ_{jk} ($p_Z(z = k | \mathbf{x}_j)$), which is then used for finding the right cluster at inference.

$$\hat{k} = \underset{k \in \{1, 2, \dots, K\}}{\operatorname{argmax}} \tau_{jk} \quad \star \star \star$$



Advantages:

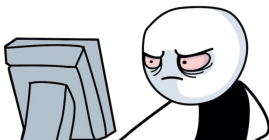
- GMMs can model clusters with various shapes depending on the assumption imposed on the covariance matrix, *i.e.*, Σ_{z_k} . Thus, it is **more flexible than K-mean clustering**.
- GMMs have **probabilistic assignment**, which gives a better overview of how likely the data can be grouped into different clusters.

Disadvantages:

- Similar to K-mean clustering, it needs a **predefined value** of the number of clusters K . Also, the initial parameter estimates can affect the final result.
- Similar to K-mean clustering, GMMs do not have a function to identify noises in data samples.
- GMMs are more **computationally intensive** than K-mean clustering.

The following slides will contain the heavy detailed information about how to estimate GMM's model parameters from Expectation-Maximization which will result in an algorithm that looks similar to the previous page.

Be prepared....



- ① Table of contents
- ② Supervised vs unsupervised learning
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What is "Expectation" Maximization?

Setting: We have

x : observed RV.

z : unknown RV.

$$\theta_{x,z} = \{ \theta_{x|z}, \theta_{z|z} \}$$

Parameter of X when Fix Z .
 Parameter of distribution Z .

ข้อมูล $X: p(x|z, \theta_{x,z})$ Observed $p(x,z, \theta_{x,z}) \cdot p(z, \theta_{x,z})$ Estimate.

เป้าหมาย $\Rightarrow \theta_{x,z}, z = \arg \max p(x|z, \theta_{x,z})$
MLE + MAP

Z ภาพทาบเทียม

noisy

X ภาพที่ได้อะไร

⊛ Target: $\hat{\theta}_{x,z} = \arg \max_{\theta_{x,z}} p(x|z, \theta_{x,z}) \triangleq \arg \max_{\theta_{x,z}} \log(p(x|z, \theta_{x,z}))$

where $p(x|z, \theta_{x,z}) = \int p(x, z | \theta_{x,z}) dz$

$$= \int \underbrace{p(x|z, \theta_{x|z})}_{\text{We assume this one}} \underbrace{p(z|\theta_z)}_{\text{But don't know this one.}}$$

เราอยากหาค่า $\theta_{x,z}$ แล้วไป marginalize z .

Then $\log p(x|\theta_{x,z}) = \log \int \underline{p(x|z) p(z)} dz$

* Dive into $p(x|z, \theta_{x|z})$ $\nwarrow$ $\theta_{x,z} = \{\theta_{x|z}, \theta_z\}$

$$p(x|z, \theta_{x,z}) = \frac{p(x, z | \theta_{x|z}, \theta_z)}{p(z | \theta_z)}$$

$$\approx \frac{p(x, z | \theta_{x|z}, \theta_z)}{p(z|x, \theta_{z|x})}$$

$\therefore$ we can only guess " z from x "
 $p(z) \approx p(z|x)$

$$\log p(x|\theta_{x,z}) = \log \int \frac{p(x, z | \theta_{x|z}, \theta_z)}{p(z|x, \theta_{z|x})} p(z) dz$$

Concave.

$$\approx \int p(z) \log \left[\frac{p(x, z | \theta_{x|z}, \theta_z)}{p(z|x, \theta_{z|x})} \right] dz$$

Problem: $p(z)$ vs. $p(z|x)$

$$\int p(z) \log \left[\frac{p(x, z | \theta_{x|z}, \theta_z)}{p(z|x, \theta_{z|x})} \right] dz$$

$$= \int p(z) \left[\log p(x, z) - \log p(z|x) \right] dz$$

$$= \int p(z) \left[\log p(x, z) - \log p(z|x) + \log p(z) - \log p(z) \right] dz$$

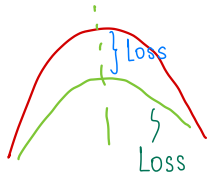
$$= \int p(z) \left[\log \frac{p(x, z)}{p(z)} - \log \frac{p(z|x)}{p(z)} \right] dz$$

$$= \underbrace{\int p(z) \log p(x|z) dz}_{\text{loss}} - \int p(z) \log \frac{p(z|x)}{p(z)} dz$$

KL, ≥ 0

+ $KL(p(z) \| p(z|x)) \leftarrow$ Bishop's Chapter 9.

$$\operatorname{argmax}_{\theta_{x,z}} \log p(x | \theta_{x,z}) \Leftrightarrow \operatorname{argmax} \text{loss} + \underbrace{KL(p(z) \| p(z|x))}_{\text{always } (+)}$$



$$\Leftrightarrow \operatorname{argmax} \text{loss}$$

$$\Leftrightarrow \operatorname{argmax}_{\theta_{x,z}} \int p(z | \theta_z) \log p(x | z, \theta_{x,z}) dz$$

Note $\int p(z | \theta_z) \log \left[\frac{p(x, z | \theta_{x,z})}{p(z | \theta_z)} \right] dz \quad \swarrow \quad p(x|z, \theta_{x|z})$

$$E_x[g(x)] = \int_x g(x) \cdot f(x) dx = E_{z \sim p(z | \theta_z)} [\log p(x | z, \theta_{x|z})]$$

$$\operatorname{argmax}_{\theta_{x,z}} \log p(x | \theta_{x,z}) \Leftrightarrow \operatorname{argmax}_{\theta_{x,z}} \underline{E_z [\log p(x | z, \theta_{x|z})]}$$

EM - Algorithm.

$$\operatorname{argmax}_{\Theta_{x,z}} E_z [\log p(x|z, \Theta_{x|z})]$$

How to use EM:

① Marginalization form.

② Find closed-form soln. of params.

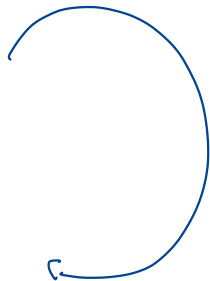
① Expectation:

Compute $\text{loss} := E_z [\log p(x|z, \hat{\Theta}_{x|z})]$

② Maximization:

Solve for $\hat{\Theta}_{x|z}$ and $\hat{\Theta}_z$ from

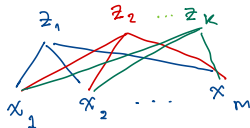
$$\hat{\Theta}_{x,y} = \operatorname{argmax}_{\Theta_{x,z}} \text{Loss}$$



EM to GMM (Multiple RVs)

$$\text{EM: } \arg \max_{\theta_{x,z}} \log p(x | \theta_{x,z}) \iff \arg \max_{z \sim p(z | \theta_z)} E \left[\log p(x | z, \theta_{x|z}) \right]$$

GMM: Training data: $\{x_j \in \mathbb{R}^N \mid j = 1, 2, \dots, m\}$
 $\xrightarrow{\text{m samples}}$



(10) x_j is assigned to Cluster k
if $k_j = \arg \max_j \tau_{jk}$

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$$\int p(z | \theta_z) \log \frac{p(x, z | \theta_{x,z})}{p(x | \theta_x)} dz$$

$$\arg \max \log p(x_1, \dots, x_m | \theta_{x_1, \dots, x_m, z_1, \dots, z_k})$$

$$\iff \arg \max_{\theta_{x_1, \dots, x_m, z_1, \dots, z_m}} \int \dots \int \log p(x_1, \dots, x_m, z_1, \dots, z_m) dz_1 \dots dz_m$$

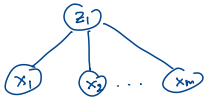
$$\iff \arg \max_{\theta_1 \dots \theta_k} \int \dots \int \log \left[\prod_{k=1}^k p(x_1, \dots, x_m | z_k) p(z_k) \right] dz_1 \dots dz_k$$

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$$\int \dots \int_{z_1 \dots z_k} \log \prod_{k=1}^K p(x_1, \dots | z_k) p(z_k) dz_1 \dots dz_k$$

$$= \int \dots \int_{z_1 \dots z_k} \left[\log p(x_1, \dots | z_1) p(z_1) + \log p(x_1, \dots | z_2) p(z_2) \dots \right]$$

$$= \int_{z_1} \log p(x_1, \dots | z_1) p(z_1) dz_1 + \int_{z_2} \log p(x_1, \dots | z_2) p(z_2) dz_2 + \dots$$



$$p(x_1, \dots, x_m | z_1) p(z_1) = \prod_{j=1}^m p(x_j | z_1) p(z_1)$$

$$= \underbrace{\int \log \prod_{j=1}^m p(x_j | z_1) p(z_1) dz_1}_{\text{IIMN}} + \underbrace{\int \log \prod_{j=1}^m p(x_j | z_2) p(z_2) dz_2}_{\text{IIMN}}$$

$$= \int \log \prod_{j=1}^m p(x_j | z_1) p(z_1) dz_1 + \int \log \prod_{j=1}^m p(x_j | z_2) p(z_2) dz_2 + \dots$$

$$= \int \sum_{j=1}^m \log p(x_j | z_1) p(z_1) dz_1 + \int \sum_{j=1}^m \log p(x_j | z_2) p(z_2) dz_2 + \dots$$

$$= \sum_{j=1}^m \int \log p(x_j | z_1) p(z_1) dz_1 + \sum_{j=1}^m \int \log p(x_j | z_2) p(z_2) dz_2 + \dots$$

* Dive into $p(x|z, \theta_{x|z})$

$$p(x|z, \theta_{x|z}) = \frac{p(x, z | \theta_{x|z}, \theta_z)}{p(z | \theta_z)}$$

$$\approx \frac{p(x, z | \theta_{x|z}, \theta_z)}{p(z | x, \theta_{z|x})}$$

$\therefore$ we can only guess " z from x "
 $p(z) \approx p(z|x)$

$$\sum_{j=1}^m \int_{z_1} \frac{\log p(x_j|z_1) p(z_1)}{p(z_1)} dz_1 + \sum_{j=1}^m \int_{z_2} \log p(x_j|z_2) p(z_2) dz_2$$

$$= \sum_{j=1}^m \int_{z_1} \log \left[\frac{p(x_j, z_1 | \theta_{x_j})}{p(z_1 | x_j)} \cdot p(z_1) \right] dz_1 + \sum_{j=1}^m \int_{z_2} \log \left[\frac{p(x_j, z_2 | \theta_{x_j})}{p(z_2 | x_j)} \cdot p(z_2) \right] dz_2$$

Concavity

$$\approx \sum_{j=1}^m \int_{z_1} p(z_1) \log \frac{p(x_j, z_1 | \theta_{x_j})}{p(z_1 | x_j)} dz_1 + \sum_{j=1}^m \int_{z_2} \frac{p(z_2)}{p(z_2 | x_j)} \log \frac{p(x_j, z_2 | \theta_{x_j})}{p(z_2 | x_j)} dz_2 + \dots$$

= Loss + KL

$$\text{Loss} = \sum_{j=1}^m \mathbb{E}_{z_1 \sim p(z_1)} \left[\log p(x_j | z_1) \right] + \sum_{j=1}^m \mathbb{E}_{z_2 \sim p(z_2)} \left[\log p(x_j | z_2) \right] + \dots$$

$$\text{Loss} = \sum_{j=1}^m \mathbb{E}_{z_1 \sim p(z_1)} \left[\log p(x_j | z_1) \right] + \sum_{j=1}^m \mathbb{E}_{z_2 \sim p(z_2)} \left[\log p(x_j | z_2) \right] + \dots$$

Each term has to be calculated independently.

So we define $\mathbb{I} \in \{0, 1\}^m$, $\mathbb{I}(j) = \begin{cases} 1 & \text{if } x_j \in C_k \\ 0 & \text{otherwise} \end{cases}$.

$$\begin{aligned} \text{Loss} &= \sum_{j=1}^m \underline{\mathbb{I}[z_j = k]} \cdot \mathbb{E}_{z \sim p(z)} \left[\log p(x_j | z) \right] \\ &= \sum_{j=1}^m \underline{\mathbb{I}[z_j = k]} \int \underbrace{p(z_k)}_{\rightarrow \approx \mathbb{I}(z_j = k)} \log p(x_j | z_k) \, dz_k \\ &= \sum_{j=1}^m \int \underline{\mathbb{I}[z_j = k] \cdot p(z_j = k)} \log p(x_j | z_j = k) \, dz_k \\ &= \sum_{j=1}^m \mathbb{E}_{z_j \sim p(z_j = k)} \left[\log p(x_j | z_j = k) \right] \end{aligned}$$

$$\text{Loss} = \sum_{j=1}^n \mathbb{E}_{z_j \sim p(z_j=k)} \left[\underline{\underline{\log p(x_j | z_j=k)}} \right]$$

$$\bullet \quad p(x_j | z_j=k) \rightarrow p_{\mathbf{X}|Z}(x_j | \mu_{z_k}, \Sigma_{z_k}) = (2\pi)^{-\frac{n}{2}} \det(\Sigma_{z_k})^{-\frac{1}{2}} \exp \left(-\frac{1}{2} (x_j - \mu_{z_k})^T \Sigma_{z_k}^{-1} (x_j - \mu_{z_k}) \right)$$

$$\log p_{\mathbf{X}|Z}(x_j | \mu_{z_k}, \Sigma_{z_k}) \propto -\frac{1}{2} \log \det(\Sigma_{z_k}) - \frac{1}{2} (x_j - \mu_{z_k})^T \Sigma_{z_k}^{-1} (x_j - \mu_{z_k})$$

$$\bullet \quad \mathbb{E}_{z_j \sim p(z_j=k)} \left[\underline{\underline{\log p(x_j | z_j=k)}} \right] = \int \overbrace{\mathbb{I}[z_j=k] \cdot p(z_j=k)}^{\mapsto \approx \mathbb{I}(z_j=k)} \log p(x_j | z_j=k) dz_k$$

$$\bullet \quad \mathbb{I}(z_j=k) \log p(x_j | z_j=k) = \mathbb{I}[z_j=k] \log p_{\mathbf{X}|Z}(x_j | \mu_{z_k}, \Sigma_{z_k})$$

$$= -\frac{n}{2} \mathbb{I}[z_j=k] \log \sigma_k^2 - \frac{1}{2\sigma_k^2} \mathbb{I}[z_j=k] (x_j - \mu_{z_k})^T (x_j - \mu_{z_k})$$

Estimating parameters using stationary conditions

Thus, we can estimate these parameters $\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K$ by solving the following maximum likelihood:

$$\underset{\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K}{\operatorname{argmax}} \quad \mathcal{L}(\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K)$$

where $\mathcal{L}(\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K)$

$$\begin{aligned} &= \sum_{j=1}^m \sum_{k=1}^K \mathbb{I}[z_j = k] \log p_{\mathbf{X}, Z}(\mathbf{x}_j, z_k | \pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}) \\ &= \sum_{j=1}^m \sum_{k=1}^K -\frac{n}{2} \mathbb{I}[z_j = k] \log \sigma_k^2 - \frac{1}{2\sigma_k^2} \mathbb{I}[z_j = k] (\mathbf{x}_j - \mu_{\mathbf{z}_k})^T (\mathbf{x}_j - \mu_{\mathbf{z}_k}) + \mathbb{I}[z_j = k] \log \pi_k \end{aligned}$$

Thus, we will estimate these parameters using stationary conditions:

1. $\nabla_{\mu_{\mathbf{z}_k}} \mathcal{L}(\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K) = 0$
 2. $\frac{d}{d\sigma_k^2} \mathcal{L}(\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K) = 0$
 3. $\frac{d}{d\pi_k} \mathcal{L}(\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K) = 0$
- } loss. $\frac{d}{d\mu_{\mathbf{z}_k}} \mathbb{I}[z_j = k]$ terms.

Note that $\mathcal{L}(\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K) = \sum_{j=1}^m \mathbb{E}_{\mathbb{I}[z_j = k]} [\log p_{\mathbf{X}, Z}(\mathbf{x}_j, z_k | \pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k})]$

Estimating the model parameters in GMM

$$\begin{aligned} 1. \quad \nabla_{\mu_{\mathbf{z}_k}} \mathcal{L}(\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K) &= 0 \\ &= -\frac{1}{2\sigma_k^2} \sum_{j \in k} \nabla_{\mu_{\mathbf{z}_k}} (\mathbf{x}_j - \mu_{\mathbf{z}_k})^T (\mathbf{x}_j - \mu_{\mathbf{z}_k}) \\ &= -\frac{1}{2\sigma_k^2} \sum_{j \in k} (-2\mathbf{x}_j + 2\mu_{\mathbf{z}_k}) \\ \Rightarrow \mu_{\mathbf{z}_k} &= \frac{1}{|k|} \sum_{j \in k} \mathbf{x}_j = \frac{\sum_{j=1}^m \mathbb{I}[z_j = k] \mathbf{x}_j}{\sum_{j=1}^m \mathbb{I}[z_j = k]} \end{aligned}$$

$$2 \frac{d}{d\sigma_k^2} \mathcal{L}(\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K) = 0$$

$$\begin{aligned} &= \sum_{j \in k} -\frac{n}{2} \frac{d}{d\sigma_k^2} (\log \sigma_k^2) - \frac{d}{d\sigma_k^2} \left(\frac{1}{2\sigma_k^2} (\mathbf{x}_j - \mu_{\mathbf{z}_k})^T (\mathbf{x}_j - \mu_{\mathbf{z}_k}) \right) \\ &= \sum_{j \in k} -\frac{n}{2} \frac{1}{\sigma_k^2} + \frac{1}{2(\sigma_k^2)^2} \left((\mathbf{x}_j - \mu_{\mathbf{z}_k})^T (\mathbf{x}_j - \mu_{\mathbf{z}_k}) \right) \\ &= \sum_{j=1}^m \left(-n \mathbb{I}[z_j = k] + \frac{\mathbb{I}[z_j = k]}{\sigma_k^2} \left((\mathbf{x}_j - \mu_{\mathbf{z}_k})^T (\mathbf{x}_j - \mu_{\mathbf{z}_k}) \right) \right) \\ \sigma_k^2 &= \frac{\sum_{j=1}^m \mathbb{I}[z_j = k] (\mathbf{x}_j - \mu_{\mathbf{z}_k})^T (\mathbf{x}_j - \mu_{\mathbf{z}_k})}{n \sum_{j=1}^m \mathbb{I}[z_j = k]} \end{aligned}$$

$$3. \quad \frac{d}{d\pi_k} \mathcal{L}(\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K) = 0$$

So, what is π_k by definition? By definition, it is the probability that a cluster exists. Thus, we can compute this term empirically:

$$\pi_k = p_Z(z = k) = \frac{\sum_{j=1}^m \mathbb{I}[z_j = k]}{m}$$

Expectation-Maximization (EM)

Let's consider a problem of the parameter estimation, i.e., Θ , from a marginal likelihood of the observed data, as follows:

$$\hat{\theta} = \underset{\theta}{\operatorname{argmax}} p_X(x|\theta) = \int_z p_{X,Z}(x, z|\theta)$$

However, it is often the case that Z is the latent variable, yet we know $p_{X|Z}(x|z)$. Instead of solving the marginalization, EM to estimate θ by iteratively refining the parameter estimation by alternating between

- **E-step:** compute the expected value of $\log p_{X,Y}(x, y, |\theta)$ with respect to $p_{Z|X}(z|x, \theta)$

$$E_{p_{Z|X}(z|x, \theta^t)} [\log p_{X|Z}(x|z, |\theta^t)] = \int_z p_{Z|X}(z|x, \theta^t) \log p_{X,Y}(x, y, |\theta^t)$$

- **M-step:** maximizing over Θ the expectation:

$$\theta^{t+1} = \underset{\theta}{\operatorname{argmax}} E_{p_{Z|X}(z|x, \theta^t)} [\log p_{X|Z}(x, y, |\theta^t)]$$

Estimating the model parameters in GMM—Summary

From our previous derivations, we can estimate the parameters

$$\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K$$

using the following formulations:

1. $\mu_{\mathbf{z}_k} = \frac{\sum_{j=1}^m \mathbb{I}[z_j=k] \mathbf{x}_j}{\sum_{j=1}^m \mathbb{I}[z_j=k]}$
2. $\Sigma_{\mathbf{z}_k} = \sigma_k^2 \mathbf{I}_{n \times n}$ where $\sigma_k^2 = \frac{\sum_{j=1}^m \mathbb{I}[z_j=k] (\mathbf{x}_j - \mu_{\mathbf{z}_k})^T (\mathbf{x}_j - \mu_{\mathbf{z}_k})}{n \sum_{j=1}^m \mathbb{I}[z_j=k]}$
3. $\pi_k = \frac{\sum_{j=1}^m \mathbb{I}[z_j=k]}{m}$

Unfortunately, we don't know both the cluster assignments ($\mathbb{I}[z_j = k]$) and the parameters ($\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K$).

To solve this problem, we will use an algorithm called **Expectation-Maximization (EM)**, which will simply guess the values of the parameters and iteratively refine these parameter estimation by alternating between **probabilistic inference** and **statistical inference**.

Applying EM to GMM parameter estimation

Before applying Expectation-Maximization onto estimating the GMM model parameters.
Previously ...

① $\mathcal{L}(\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K) = \sum_{j=1}^m \mathbb{E}_{\mathbb{I}[z_j=k]} [\log p_{\mathbf{X},Z}(\mathbf{x}_j, z_k | \pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k})].$

Thus, we can directly apply M-step to update the model parameter.

- ② The cluster assignment $\mathbb{I}[z_j = k]$ maps each training sample $j = 1, \dots, m$ to the k th cluster: for all $j = 1, 2, \dots, m$

$$\mathbb{I}[z_j = k] = \begin{cases} 1 & \text{if } j \in k \\ 0 & \text{otherwise} \end{cases}$$

This behaves similar to $p_{Z|X}(z = k | \mathbf{x}_j)$ or τ_{jk} .

- ③ Therefore, one can estimate $\mathbb{I}[z_j = k] \approx p_{Z|X}(z = k | \mathbf{x}_j)$ or τ_{jk} .

$$\mathbb{I}[z_j = k] = p_{Z|X}(z = k | \mathbf{x}_j) = \frac{\pi_k p_{\mathbf{X}|Z}(\mathbf{x}_j | \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k})}{p_{\mathbf{X}}(\mathbf{x}_j)} \propto \pi_k p_{\mathbf{X}|Z}(\mathbf{x}_j | \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k})$$

GMM parameter estimation using EM — the algorithm

We can apply EM to the GMM parameter estimation as follows:

Predefine the number of clusters K and initialize $\pi_k^t, \mu_{\mathbf{z}_k}^t, \Sigma_k^t$.

- **E-step:** Update $\mathbb{I}^{t+1}[z_j = k]$ (or τ_{jk}) as follows:

$$\mathbb{I}^{t+1}[z_j = k] = \pi_k^t p_{\mathbf{X}|Z}(\mathbf{x}_j | \mu_{\mathbf{z}_k}^t, \Sigma_k^t)$$

Then, compute

$$\sum_{j=1}^m \mathbb{E}_{\mathbb{I}^{t+1}[z_j=k]} [\log p_{\mathbf{X}|Z}(\mathbf{x}_j, z_k | \pi_k^t, \mu_{\mathbf{z}_k}^t, \Sigma_{\mathbf{z}_k}^t)]$$

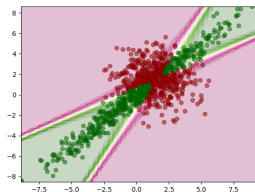
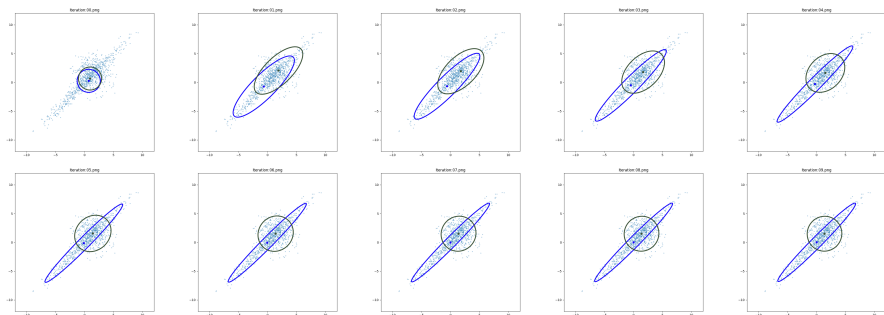
- **M-step:** updating the GMM parameters. Given $\mathbb{I}^{t+1}[z_j = k]$

$$\{\pi_k^{t+1}, \mu_{\mathbf{z}_k}^{t+1}, \Sigma_{\mathbf{z}_k}^{t+1}\}_{k=1}^K = \underset{\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K}{\operatorname{argmax}} \sum_{j=1}^m \mathbb{E}_{\mathbb{I}^{t+1}[z_j=k]} [\log p_{\mathbf{X}|Z}(\mathbf{x}_j, z_k | \pi_k^t, \mu_{\mathbf{z}_k}^t, \Sigma_{\mathbf{z}_k}^t)]$$

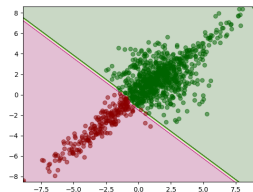
where each parameters can be updated as follows:

$$\begin{aligned}\pi_k^{t+1} &= \frac{1}{m} \sum_{j=1}^m \mathbb{I}^{t+1}[z_j = k] \\ \mu_{\mathbf{z}_k}^{t+1} &= \frac{\sum_{j=1}^m \mathbb{I}^{t+1}[z_j = k] \mathbf{x}_j}{\sum_{j=1}^m \mathbb{I}^{t+1}[z_j = k]} \\ \Sigma_{\mathbf{z}_k}^{t+1} &= \frac{\sum_{j=1}^m \mathbb{I}^{t+1}[z_j = k] (\mathbf{x}_j - \mu_k)(\mathbf{x}_j - \mu_k)^T}{\sum_{j=1}^m \mathbb{I}^{t+1}[z_j = k]}\end{aligned}$$

Expectation-Maximization for GMM — the result



GMM



K-Mean

Training and inference phase of GMM clustering

- **Training.** On the training dataset, learn the GMM model parameters by solving the following maximum likelihood:

$$\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K = \underset{\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K}{\operatorname{argmax}} \sum_{j=1}^m \sum_{k=1}^K \mathbb{I}[z_j = k] \log p_{\mathbf{X}, Z}(\mathbf{x}_j, z_k | \pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k})$$

However, because the cluster assignment $\mathbb{I}[z_j = k]$ is unknown, therefore, the above optimization problem is solved via an expectation maximization (EM) algorithm.

- **Inference.** Given the input testing sample $\mathbf{x}$, use the learned GMM model to compute $p_{Z|\mathbf{X}}(z = k|\mathbf{x})$. Then, one can find the cluster for a testing sample $\mathbf{x}$ by

$$\hat{k} = \underset{k \in 1, 2, \dots, K}{\operatorname{argmax}} p_{Z|\mathbf{X}}(z = k|\mathbf{x})$$

Coding for EM in GMM clustering

```
def e_step(self):  
    """  
    E-step of EM algorithm.  
    """  
    for i in range(self.k):  
        self.z[:, i] = self.pi[i] * multivariate_normal.pdf(self.data, mean=self.mu[i], cov=self.sigma[i])  
    self.z /= self.z.sum(axis=1, keepdims=True)  
  
def m_step(self):  
    """  
    M-step of EM algorithm.  
    """  
    sum_z = self.z.sum(axis=0)  
    self.pi = sum_z / self.num_points  
    self.mu = np.matmul(self.z.T, self.data)  
    self.mu /= sum_z[:, None]  
    for i in range(self.k):  
        j = np.expand_dims(self.data, axis=1) - self.mu[i]  
        s = np.matmul(j.transpose([0, 2, 1]), j)  
        self.sigma[i] = np.matmul(s.transpose(1, 2, 0), self.z[:, i])  
        self.sigma[i] /= sum_z[i]
```

Reference: [EM implementation](#)

- [1] Christopher M. Bishop. *Pattern Recognition and Machine Learning (Information Science and Statistics)*. Berlin, Heidelberg: Springer-Verlag, 2006. ISBN: 0387310738.
- [2] Gareth James et al. *An Introduction to Statistical Learning: with Applications in R*. Springer, 2013. URL: https://hastie.su.domains/ISLP/ISLP_website.pdf.download.html.
- [3] Lester Mackey. *Lecture 2 Gaussian mixture model*. in STATS 306B, Unsupervised Learning, Spring 2014. URL: <http://web.stanford.edu/class/stats200/Lecture14.pdf>.

Settings of the parameter estimation in GMMs

Let's start with how to estimate these parameters $\{\pi_k, \mu_{z_k}, \Sigma_{z_k}\}_{k=1}^K$.

We can estimate these parameters by solving the following maximum likelihood:

$$\begin{aligned} \underset{\{\pi_k, \mu_{z_k}, \Sigma_{z_k}\}_{k=1}^K}{\operatorname{argmax}} \quad & \mathcal{L}(\{\pi_k, \mu_{z_k}, \Sigma_{z_k}\}_{k=1}^K) \quad \text{*** also can EM method derive} \\ & \text{an Joint answer.} \\ \mathcal{L}(\{\pi_k, \mu_{z_k}, \Sigma_{z_k}\}_{k=1}^K) = & \prod_{k=1}^K \prod_{j \in k} p_{\mathbf{X}, Z}(\mathbf{x}_j, z_k | \pi_k, \mu_{z_k}, \Sigma_{z_k}) \\ & \downarrow \text{I'll abuse step 1b} \\ \propto & \sum_{k=1}^K \sum_{j \in k} \log p_{\mathbf{X}, Z}(\mathbf{x}_j, z_k | \pi_k, \mu_{z_k}, \Sigma_{z_k}) \\ = & \sum_{j=1}^m \sum_{k=1}^K \mathbb{I}[z_j = k] \log p_{\mathbf{X}, Z}(\mathbf{x}_j, z_k | \pi_k, \mu_{z_k}, \Sigma_{z_k}) \\ \left\{ \right. = & \sum_{j=1}^m \mathbb{E}_{\mathbb{I}[z_j = k]} [\log p_{\mathbf{X}, Z}(\mathbf{x}_j, z_k | \pi_k, \mu_{z_k}, \Sigma_{z_k})] \end{aligned}$$

where $\mathbb{I}[z_j = k]$ refers to as the cluster assignments, which will take a value '1', if x_j belong to the k cluster. Also, we choose $p_{\mathbf{X}, Z}(\mathbf{x}_j, z = k)$ instead of $p_{\mathbf{X}}(\mathbf{x}_j)$, because $p_{\mathbf{X}}(\mathbf{x}_j)$ requires the marginalization of the latent variable z , which causes many unknowns to couple together.

Next we will focus on decomposing $\mathbb{I}[z_j = k] \log p_{\mathbf{X}, Z}(\mathbf{x}_j, z_k | \pi_k, \mu_{z_k}, \Sigma_{z_k})$

Estimating the model parameters in GMM

$$\mathbb{I}[z_j = k] \log p_{\mathbf{X}, Z}(\mathbf{x}_j, z_k | \pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}) = \mathbb{I}[z_j = k] \log p_{\mathbf{X} | Z}(\mathbf{x}_j | \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}) + \mathbb{I}[z_j = k] \log \pi_k$$

For $p_{\mathbf{X} | Z}(\mathbf{x}_j | \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k})$ being a Gaussian, we have that :

$$p_{\mathbf{X} | Z}(\mathbf{x}_j | \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}) = (2\pi)^{-\frac{n}{2}} \det(\Sigma_{\mathbf{z}_k})^{-\frac{1}{2}} \exp \left(-\frac{1}{2} (\mathbf{x}_j - \mu_{\mathbf{z}_k})^T \Sigma_{\mathbf{z}_k}^{-1} (\mathbf{x}_j - \mu_{\mathbf{z}_k}) \right)$$

$$\log p_{\mathbf{X} | Z}(\mathbf{x}_j | \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}) \propto -\frac{1}{2} \log \det(\Sigma_{\mathbf{z}_k}) - \frac{1}{2} (\mathbf{x}_j - \mu_{\mathbf{z}_k})^T \Sigma_{\mathbf{z}_k}^{-1} (\mathbf{x}_j - \mu_{\mathbf{z}_k})$$

For simplicity, assume that the data in feature dimension $\mathcal{R}^n$ in each $\mathbf{x}_j$ are uncorrelated and distributed with the same variance, if they belong to the same cluster: $\Sigma_{\mathbf{z}_k} = \sigma_k^2 \mathbf{I}_{n \times n}$.

$$\log p_{\mathbf{X} | Z}(\mathbf{x}_j | \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}) \propto -\frac{n}{2} \log(\sigma_k^2) - \frac{1}{2\sigma_k^2} (\mathbf{x}_j - \mu_{\mathbf{z}_k})^T (\mathbf{x}_j - \mu_{\mathbf{z}_k})$$

Therefore, we can decompose $\mathbb{I}[z_j = k] \log p_{\mathbf{X}, Z}(\mathbf{x}_j, z_k | \pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k})$ as follows:

$$\mathbb{I}[z_j = k] \log p_{\mathbf{X}, Z}(\mathbf{x}_j, z_k | \pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k})$$

$$= \mathbb{I}[z_j = k] \log p_{\mathbf{X} | Z}(\mathbf{x}_j | \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}) + \mathbb{I}[z_j = k] \log \pi_k$$

$$= -\frac{n}{2} \mathbb{I}[z_j = k] \log \sigma_k^2 - \frac{1}{2\sigma_k^2} \mathbb{I}[z_j = k] (\mathbf{x}_j - \mu_{\mathbf{z}_k})^T (\mathbf{x}_j - \mu_{\mathbf{z}_k}) + \mathbb{I}[z_j = k] \log \pi_k$$

Estimating the model parameters in GMM

$$3 \quad \frac{d}{d\pi_k} \mathcal{L}(\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K) = 0$$
$$= \sum_{j \in k} \frac{d}{d\pi_k} \log \pi_k = \sum_{j \in k} \frac{1}{\pi_k} = \frac{|k|}{\pi_k} = 0$$

This will be true if $\pi_k \rightarrow \infty$. However, this is impossible as $\pi_k \in [0, 1]$ and $\sum_{k=1}^K \pi_k = 1$. Let's consider these constraints together with the loss function via Lagrangian multiplier:

$$\mathcal{LL} = \mathcal{L}(\{\pi_k, \mu_{\mathbf{z}_k}, \Sigma_{\mathbf{z}_k}\}_{k=1}^K) - \lambda \left(\sum_{j=1}^m \sum_{k=1}^K \pi_k - 1 \right) - \beta \sum_{j=1}^m \sum_{k=1}^K \pi_k - \gamma \sum_{j=1}^m \sum_{k=1}^K (1 - \pi_k)$$

$$\Rightarrow \frac{d}{d\pi_k} \mathcal{LL} = \frac{|k|}{\pi_k} - \lambda |k| - \beta |k| + \gamma |k| = 0$$

$$\pi_k = \frac{1}{\lambda + \beta - \gamma} \text{ which means that } \pi_k \text{ can be any real value } \in (0, 1)$$

So, what is π_k by definition? By definition, it is the probability that a cluster exists. Thus, we can compute this term empirically:

$$\pi_k = p_Z(z = k) = \frac{\sum_{j=1}^m \mathbb{I}[z_j = k]}{m}$$